confidence_performance_perception_uncertainty_score_SCR

October 9, 2026

1 Multilevel Analysis for Retrospective Confidence, Perceived Individual Performance, and Team Gameplay Outcome based on Skin Conductance Response

by Elvis Ortega-Ochoa

[1]:
```
import sys
import subprocess

subprocess.check_call([sys.executable, "-m", "pip", "install", "--upgrade",␣
 ↪"pip", "ipywidgets", "jupyterlab_widgets"])
subprocess.check_call([
    sys.executable, "-m", "pip", "install",
    "pymer4",
    "rpy2",
    "polars",
    "great-tables",
    "joblib",
    "scikit-learn",
    "pyarrow",
    "bambi",
    "arviz",
    "matplotlib",
    "pandas",
    "numpy",
    "statsmodels",
    "openpyxl",
])
```

```
Requirement already satisfied: pip in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(26.2.1)
Requirement already satisfied: ipywidgets in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(8.1.9)
Requirement already satisfied: jupyterlab_widgets in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(3.0.17)
```

```
Requirement already satisfied: comm>=0.1.3 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from ipywidgets) (0.2.3)
Requirement already satisfied: ipython>=6.1.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from ipywidgets) (9.11.0)
Requirement already satisfied: traitlets>=4.3.1 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from ipywidgets) (5.14.3)
Requirement already satisfied: widgetsnbextension~=4.0.16 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from ipywidgets) (4.0.16)
Requirement already satisfied: decorator>=5.1.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from ipython>=6.1.0->ipywidgets) (5.2.1)
Requirement already satisfied: ipython-pygments-lexers>=1.0.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from ipython>=6.1.0->ipywidgets) (1.1.1)
Requirement already satisfied: jedi>=0.18.2 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from ipython>=6.1.0->ipywidgets) (0.19.2)
Requirement already satisfied: matplotlib-inline>=0.1.6 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from ipython>=6.1.0->ipywidgets) (0.2.1)
Requirement already satisfied: pexpect>4.6 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from ipython>=6.1.0->ipywidgets) (4.9.0)
Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from ipython>=6.1.0->ipywidgets) (3.0.52)
Requirement already satisfied: pygments>=2.14.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from ipython>=6.1.0->ipywidgets) (2.19.2)
Requirement already satisfied: stack_data>=0.6.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from ipython>=6.1.0->ipywidgets) (0.6.3)
Requirement already satisfied: wcwidth in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from prompt_toolkit<3.1.0,>=3.0.41->ipython>=6.1.0->ipywidgets) (0.6.0)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from jedi>=0.18.2->ipython>=6.1.0->ipywidgets) (0.8.6)
Requirement already satisfied: ptyprocess>=0.5 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from pexpect>4.6->ipython>=6.1.0->ipywidgets) (0.7.0)
Requirement already satisfied: executing>=1.2.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from stack_data>=0.6.0->ipython>=6.1.0->ipywidgets) (2.2.1)
```

```
Requirement already satisfied: asttokens>=2.1.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from stack_data>=0.6.0->ipython>=6.1.0->ipywidgets) (3.0.1)
Requirement already satisfied: pure-eval in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from stack_data>=0.6.0->ipython>=6.1.0->ipywidgets) (0.2.3)
Requirement already satisfied: pymer4 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(0.9.2)
Requirement already satisfied: rpy2 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(3.6.7)
Requirement already satisfied: polars in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(1.41.0)
Requirement already satisfied: great-tables in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(0.21.0)
Requirement already satisfied: joblib in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(1.5.3)
Requirement already satisfied: scikit-learn in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(1.8.0)
Requirement already satisfied: pyarrow in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(24.0.0)
Requirement already satisfied: bambi in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(0.17.2)
Requirement already satisfied: arviz in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(0.23.4)
Requirement already satisfied: matplotlib in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(3.10.8)
Requirement already satisfied: pandas in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(3.0.1)
Requirement already satisfied: numpy in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(2.4.3)
Requirement already satisfied: statsmodels in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(0.14.6)
Requirement already satisfied: openpyxl in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(3.1.5)
```

```
Requirement already satisfied: rpy2-rinterface>=3.6.6 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from rpy2) (3.6.6)
Requirement already satisfied: rpy2-robjects>=3.6.5 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from rpy2) (3.6.5)
Requirement already satisfied: polars-runtime-32==1.41.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from polars) (1.41.0)
Requirement already satisfied: commonmark>=0.9.1 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from great-tables) (0.9.1)
Requirement already satisfied: faicons>=0.2.2 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from great-tables) (0.2.2)
Requirement already satisfied: htmltools>=0.4.1 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from great-tables) (0.7.0)
Requirement already satisfied: importlib-metadata in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from great-tables) (9.0.0)
Requirement already satisfied: typing_extensions>=3.10.0.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from great-tables) (4.15.0)
Requirement already satisfied: Babel>=2.13.1 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from great-tables) (2.18.0)
Requirement already satisfied: importlib-resources in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from great-tables) (7.1.0)
Requirement already satisfied: scipy>=1.10.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from scikit-learn) (1.17.1)
Requirement already satisfied: threadpoolctl>=3.2.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from scikit-learn) (3.6.0)
Requirement already satisfied: arviz_plots<1,>=0.8.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from bambi) (0.8.0)
Requirement already satisfied: formulae<0.7.0,>=0.6.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from bambi) (0.6.2)
Requirement already satisfied: graphviz in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from bambi) (0.21)
Requirement already satisfied: pymc<6,>=5.27.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from bambi) (5.28.1)
```

```
Requirement already satisfied: sparse<0.18,>=0.17.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from bambi) (0.17.0)
Requirement already satisfied: arviz-base<0.9.0,>=0.8.1 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from arviz_plots<1,>=0.8.0->bambi) (0.8.2)
Requirement already satisfied: arviz-stats<0.9.0,>=0.8.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from arviz-stats[xarray]<0.9.0,>=0.8.0->arviz_plots<1,>=0.8.0->bambi) (0.8.0)
Requirement already satisfied: xarray>=2024.11.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from arviz-base<0.9.0,>=0.8.1->arviz_plots<1,>=0.8.0->bambi) (2026.2.0)
Requirement already satisfied: xarray-einstats in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from arviz-stats[xarray]<0.9.0,>=0.8.0->arviz_plots<1,>=0.8.0->bambi) (0.10.0)
Requirement already satisfied: packaging in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from formulae<0.7.0,>=0.6.0->bambi) (26.0)
Requirement already satisfied: cachetools<7,>=4.2.1 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from pymc<6,>=5.27.0->bambi) (6.2.6)
Requirement already satisfied: cloudpickle in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from pymc<6,>=5.27.0->bambi) (3.1.2)
Requirement already satisfied: pytensor<2.39,>=2.38.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from pymc<6,>=5.27.0->bambi) (2.38.2)
Requirement already satisfied: rich>=13.7.1 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from pymc<6,>=5.27.0->bambi) (14.3.3)
Requirement already satisfied: setuptools>=60.0.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from arviz) (82.0.1)
Requirement already satisfied: h5netcdf>=1.0.2 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from arviz) (1.8.1)
Requirement already satisfied: h5py in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from arviz) (3.16.0)
Requirement already satisfied: platformdirs in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from arviz) (4.9.4)
Requirement already satisfied: numba<1,>0.57 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from pytensor<2.39,>=2.38.0->pymc<6,>=5.27.0->bambi) (0.64.0)
Requirement already satisfied: filelock>=3.15 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from pytensor<2.39,>=2.38.0->pymc<6,>=5.27.0->bambi) (3.25.2)
```

```
Requirement already satisfied: etuples in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from pytensor<2.39,>=2.38.0->pymc<6,>=5.27.0->bambi) (0.3.10)
Requirement already satisfied: logical-unification in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from pytensor<2.39,>=2.38.0->pymc<6,>=5.27.0->bambi) (0.4.7)
Requirement already satisfied: miniKanren in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from pytensor<2.39,>=2.38.0->pymc<6,>=5.27.0->bambi) (1.0.5)
Requirement already satisfied: cons in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from pytensor<2.39,>=2.38.0->pymc<6,>=5.27.0->bambi) (0.4.7)
Requirement already satisfied: llvmlite<0.47,>=0.46.0dev0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from numba<1,>0.57->pytensor<2.39,>=2.38.0->pymc<6,>=5.27.0->bambi) (0.46.0)
Requirement already satisfied: contourpy>=1.0.1 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from matplotlib) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from matplotlib) (1.5.0)
Requirement already satisfied: pillow>=8 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from matplotlib) (12.1.1)
Requirement already satisfied: pyparsing>=3 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from matplotlib) (2.9.0.post0)
Requirement already satisfied: patsy>=0.5.6 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from statsmodels) (1.0.2)
Requirement already satisfied: et-xmlfile in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from openpyxl) (2.0.0)
Requirement already satisfied: six>=1.5 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from python-dateutil>=2.7->matplotlib) (1.17.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from rich>=13.7.1->pymc<6,>=5.27.0->bambi) (4.0.0)
```

```
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from rich>=13.7.1->pymc<6,>=5.27.0->bambi) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from markdown-it-py>=2.2.0->rich>=13.7.1->pymc<6,>=5.27.0->bambi) (0.1.2)
Requirement already satisfied: cffi>=1.15.1 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from rpy2-rinterface>=3.6.6->rpy2) (2.0.0)
Requirement already satisfied: pycparser in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from cffi>=1.15.1->rpy2-rinterface>=3.6.6->rpy2) (3.0)
Requirement already satisfied: jinja2 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from rpy2-robjects>=3.6.5->rpy2) (3.1.6)
Requirement already satisfied: tzlocal in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from rpy2-robjects>=3.6.5->rpy2) (5.3.1)
Requirement already satisfied: toolz in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from logical-unification->pytensor<2.39,>=2.38.0->pymc<6,>=5.27.0->bambi)
(1.1.0)
Requirement already satisfied: multipledispatch in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from logical-unification->pytensor<2.39,>=2.38.0->pymc<6,>=5.27.0->bambi)
(1.0.0)
Requirement already satisfied: zipp>=3.20 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from importlib-metadata->great-tables) (4.1.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(from jinja2->rpy2-robjects>=3.6.5->rpy2) (3.0.3)
```

[1]:
```
0
```

Packages

[55]:
```
import pandas as pd
import numpy as np
import bambi as bmb
import arviz as az
import matplotlib.pyplot as plt
import statsmodels.formula.api as smf
import matplotlib.patches as patches
import statsmodels.api as sm
import pymer4.models as pm
import polars as pl
import ipywidgets as widgets
```

```
from pymer4.models import lmer as Lmer
```

Loading data

[3]:
```
# Reading files and printing columns
scr = pd.read_excel("../../confidence_SCR/stable_scr_events_UPD.xlsx",␣
 ↪keep_default_na=False)
cf = pd.read_excel("../../confidence_SCR/CONT_dependent_variables_confidence_E.
 ↪xlsx", keep_default_na=False)

# Make a clean copy
scr = scr.copy()
cf = cf.copy()

# Extract participantID and sensorID from scrID
scr["participantID"] = scr["scrID"].str.extract(r"SCR_([^.]+)\.")
scr["sensorID"] = scr["scrID"].str.extract(r"\.([A-Z]{2})-EDA$")

# Merge the dataframes on groupID, participantID, and sensorID
dataset = scr.merge(
    cf,
    on=["groupID", "participantID", "sensorID"],
    how="inner"
)

# Rename columns for clarity
dataset = dataset.rename(columns={
    "realGamePeaks/m": "realGamePeaks_per_m",
    "realTutoPeaks/m": "realTutoPeaks_per_m",
    "realQuestionsPeaks/m": "realQuestionsPeaks_per_m",
    "realPitchPeaks/m": "realPitchPeaks_per_m",
    "CF1": "cf",
    "PP1": "pp"
})
```

Build questionPath as numeric

[4]:
```
# Build numeric questionPath at group level
dataset["qBonusPeaks"] = pd.to_numeric(dataset["qBonusPeaks"], errors="coerce")
dataset["qPenaltyPeaks"] = pd.to_numeric(dataset["qPenaltyPeaks"],␣
 ↪errors="coerce")

dataset["has_bonus"] = dataset["qBonusPeaks"].notna()
dataset["has_penalty"] = dataset["qPenaltyPeaks"].notna()

group_path = (
    dataset.groupby("groupID")[["has_bonus", "has_penalty"]]
    .any()
```

```
    .reset_index()
)

group_path["questionPath"] = np.select(
    [
        group_path["has_bonus"] & ~group_path["has_penalty"],
        group_path["has_penalty"] & ~group_path["has_bonus"],
    ],
    [1, 0],
    default=np.nan,
)

# Drop any existing questionPath variants to avoid suffixes, then merge
dataset = dataset.drop(columns=[c for c in dataset.columns if c.
 ↪startswith("questionPath")], errors="ignore")
dataset = dataset.merge(group_path[["groupID", "questionPath"]], on="groupID",␣
 ↪how="left")
```

Reverse team gameplay score

[5]:
```
# Build numeric uncertScore
dataset["uncertScore"] = pd.to_numeric(dataset["uncertScore"], errors="coerce")

# Reverse-coding, using the instrument's theoretical bounds, not the sample's
SCALE_MIN, SCALE_MAX = 1, 70  # instrument range, per ¶102
dataset["uncertScore"] = (SCALE_MIN + SCALE_MAX) - dataset["uncertScore"]

# Reverse-score uncertScore using the observed scale range
#dataset["uncertScore"] = dataset["uncertScore"].max() + dataset["uncertScore"].
 ↪min() - dataset["uncertScore"]
```

[6]:
```
# Finalize data types
dataset["groupID"] = dataset["groupID"].astype("category")
dataset["cf"] = pd.to_numeric(dataset["cf"], errors="coerce")
dataset["questionPath"] = pd.Categorical(dataset["questionPath"],␣
 ↪categories=[0,1])
```

[7]:
```
# Keep just variables needed
dat_model = dataset[["groupID", "participantID", "cf", "pp", "uncertScore",␣
 ↪"realGamePeaks_per_m", "realTutoPeaks_per_m", "realQuestionsPeaks_per_m",␣
 ↪"realPitchPeaks_per_m", "questionPath", "pitchRole"]].copy()
```

1.1 Mixed Linear Models

1.1.1 Within/Between Descomposition

[58]:
```
def add_within_between(df, raw_col, prefix):
    group_mean = df.groupby("groupID")[raw_col].transform("mean")
    within = df[raw_col] - group_mean
    between = group_mean - group_mean.mean()

    df[f"{prefix}_group_mean"] = group_mean
    df[f"{prefix}_within"] = within
    df[f"{prefix}_between"] = between

    return df

dat_model = add_within_between(dat_model, "realGamePeaks_per_m", "scr")
dat_model = add_within_between(dat_model, "realTutoPeaks_per_m", "tuto")
dat_model = add_within_between(dat_model, "realQuestionsPeaks_per_m",␣
 ↪"questions")
dat_model = add_within_between(dat_model, "realPitchPeaks_per_m", "pitch")
dat_model = add_within_between(dat_model, "pp", "pp")
dat_model = add_within_between(dat_model, "cf", "cf")
```

1.1.2 Standarize predictor variables

[64]:
```
dat_model["realGamePeaks_per_m_z"] = (
    dat_model["realGamePeaks_per_m"] - dat_model["realGamePeaks_per_m"].mean()
) / dat_model["realGamePeaks_per_m"].std()

dat_model["scr_within_z"] = (
    dat_model["scr_within"] - dat_model["scr_within"].mean()
) / dat_model["scr_within"].std()

dat_model["scr_between_z"] = (
    dat_model["scr_between"] - dat_model["scr_between"].mean()
) / dat_model["scr_between"].std()

dat_model["tuto_within_z"] = (
    dat_model["tuto_within"] - dat_model["tuto_within"].mean()
) / dat_model["tuto_within"].std()

dat_model["tuto_between_z"] = (
    dat_model["tuto_between"] - dat_model["tuto_between"].mean()
) / dat_model["tuto_between"].std()

dat_model["questions_within_z"] = (
    dat_model["questions_within"] - dat_model["questions_within"].mean()
) / dat_model["questions_within"].std()
```

```
dat_model["questions_between_z"] = (
    dat_model["questions_between"] - dat_model["questions_between"].mean()
) / dat_model["questions_between"].std()

dat_model["pitch_within_z"] = (
    dat_model["pitch_within"] - dat_model["pitch_within"].mean()
) / dat_model["pitch_within"].std()

dat_model["pitch_between_z"] = (
    dat_model["pitch_between"] - dat_model["pitch_between"].mean()
) / dat_model["pitch_between"].std()

dat_model["pp_z"] = (
    dat_model["pp"] - dat_model["pp"].mean()
) / dat_model["pp"].std()

dat_model["pp_within_z"] = (
    dat_model["pp_within"] - dat_model["pp_within"].mean()
) / dat_model["pp_within"].std()

dat_model["pp_between_z"] = (
    dat_model["pp_between"] - dat_model["pp_between"].mean()
) / dat_model["pp_between"].std()

dat_model["uncertScore_z"] = (
    dat_model["uncertScore"] - dat_model["uncertScore"].mean()
) / dat_model["uncertScore"].std()

dat_model["cf_z"] = (
    dat_model["cf"] - dat_model["cf"].mean()
) / dat_model["cf"].std()

dat_model["cf_within_z"] = (
    dat_model["cf_within"] - dat_model["cf_within"].mean()
) / dat_model["cf_within"].std()

dat_model["cf_between_z"] = (
    dat_model["cf_between"] - dat_model["cf_between"].mean()
) / dat_model["cf_between"].std()

dat_model.columns.tolist()

# Team-level dataset (one row per team, n = 29) for team gameplay score models
team_model = dat_model.drop_duplicates(subset="groupID").copy()
```

Pitch data cleaning

[67]:
```
dat_model['pitchRole'] = pd.to_numeric(dat_model['pitchRole'], errors='coerce')
dat_model_pitch_role = dat_model.dropna(subset=['groupID',
 'participantID',
 'cf',
 'pp',
 'uncertScore',
 'realGamePeaks_per_m',
 'realTutoPeaks_per_m',
 'realQuestionsPeaks_per_m',
 'realPitchPeaks_per_m',
 'questionPath',
 'pitchRole',
 'scr_group_mean',
 'scr_within',
 'scr_between',
 'tuto_group_mean',
 'tuto_within',
 'tuto_between',
 'questions_group_mean',
 'questions_within',
 'questions_between',
 'pitch_group_mean',
 'pitch_within',
 'pitch_between',
 'pp_group_mean',
 'pp_within',
 'pp_between',
 'realGamePeaks_per_m_z',
 'scr_within_z',
 'scr_between_z',
 'tuto_within_z',
 'tuto_between_z',
 'questions_within_z',
 'questions_between_z',
 'pitch_within_z',
 'pitch_between_z',
 'pp_z',
 'pp_within_z',
 'pp_between_z',
 'uncertScore_z',
 'cf_z',
 'cf_within_z',
 'cf_between_z']).copy()

dat_model_pitch_role = dat_model_pitch_role[dat_model_pitch_role['pitchRole'].
 ↪isin([0, 1])].copy()
```

```
# Finalize data type
dat_model_pitch_role["pitchRole"] = pd.
 ↪Categorical(dat_model_pitch_role["pitchRole"], categories=[0, 1])
```

1.2 Exploratory Data Analysis

[22]:
```
cols = ["cf", "pp", "uncertScore", "realGamePeaks_per_m",␣
 ↪"realTutoPeaks_per_m", "realQuestionsPeaks_per_m", "realPitchPeaks_per_m"]

# Labels to display
labels = {
    "cf": "Retrospective confidence",
    "pp": "Perceived individual performance",
    "pp_within": "Within-team PIP",
    "pp_between": "Between-team PIP",
    "uncertScore": "Team gameplay outcome",
    "realGamePeaks_per_m": "The whole GLS NS-SCR frequency",
    "scr_within": "Within-team the whole GLS NS-SCR frequency",
    "scr_between": "Between-team the whole GLS NS-SCR frequency",
    "realTutoPeaks_per_m": "Tutorial NS-SCR frequency",
    "tuto_within": "Within-team tutorial NS-SCR frequency",
    "tuto_between": "Between-team tutorial NS-SCR frequency",
    "realQuestionsPeaks_per_m": "Gameplay NS-SCR frequency",
    "questions_within": "Within-team gameplay NS-SCR frequency",
    "questions_between": "Between-team gameplay NS-SCR frequency",
    "realPitchPeaks_per_m": "Pitch NS-SCR frequency",
    "pitch_within": "Within-team pitch NS-SCR frequency",
    "pitch_between": "Between-team pitch NS-SCR frequency"
}

# Basic summary
print("Missing values:")
print(dat_model[cols].isna().sum(), "\n")

print("Descriptive statistics:")
display(dat_model[cols].describe().T)

# Distribution plots and boxplots
fig, axes = plt.subplots(2, len(cols), figsize=(20, 8))

for i, col in enumerate(cols):
    axes[0, i].hist(dat_model[col], bins=12, edgecolor="black")
    axes[0, i].set_title(f"Histogram: {col}")
    axes[0, i].set_xlabel(col)
    axes[0, i].set_ylabel("Count")

    axes[1, i].boxplot(dat_model[col], vert=True)
```

```
    axes[1, i].set_title(f"Boxplot: {col}")
    axes[1, i].set_xticks([])

plt.tight_layout()
plt.show()

# Correlation matrix
corr = dat_model[cols].corr()

fig, ax = plt.subplots(figsize=(8, 6))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)

display_labels = [labels[col] for col in cols]

ax.set_xticks(range(len(display_labels)))
ax.set_yticks(range(len(display_labels)))
ax.set_xticklabels(display_labels, rotation=45, ha="right")
ax.set_yticklabels(display_labels)
ax.set_title("General Correlation Matrix")

# Add correlation scores in each cell
for i in range(len(cols)):
    for j in range(len(cols)):
        value = corr.iloc[i, j]
        text_color = "white" if abs(value) > 0.5 else "black"
        ax.text(j, i, f"{value:.2f}", ha="center", va="center",␣
 ↪color=text_color)

plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
plt.tight_layout()
plt.show()
```

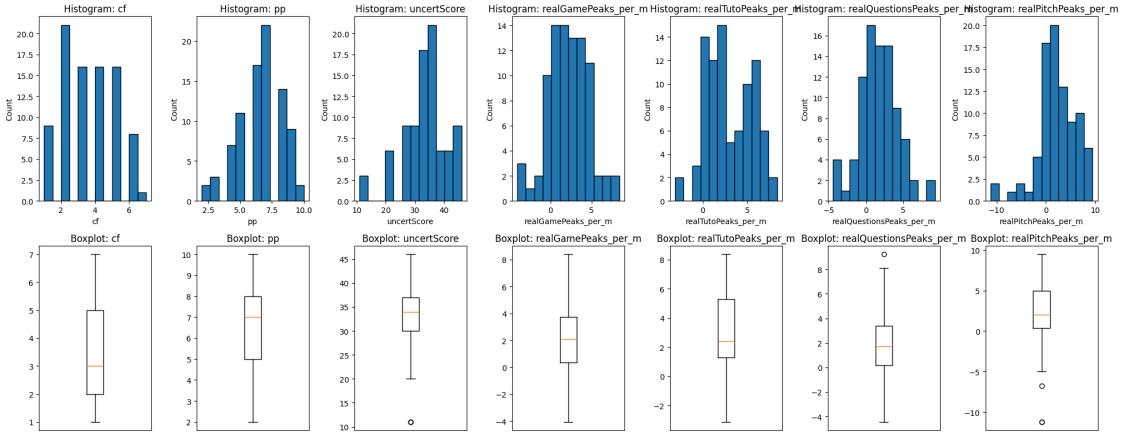
```
Missing values:
cf                          0
pp                          0
uncertScore                 0
realGamePeaks_per_m         0
realTutoPeaks_per_m         0
realQuestionsPeaks_per_m    0
realPitchPeaks_per_m        0
dtype: int64

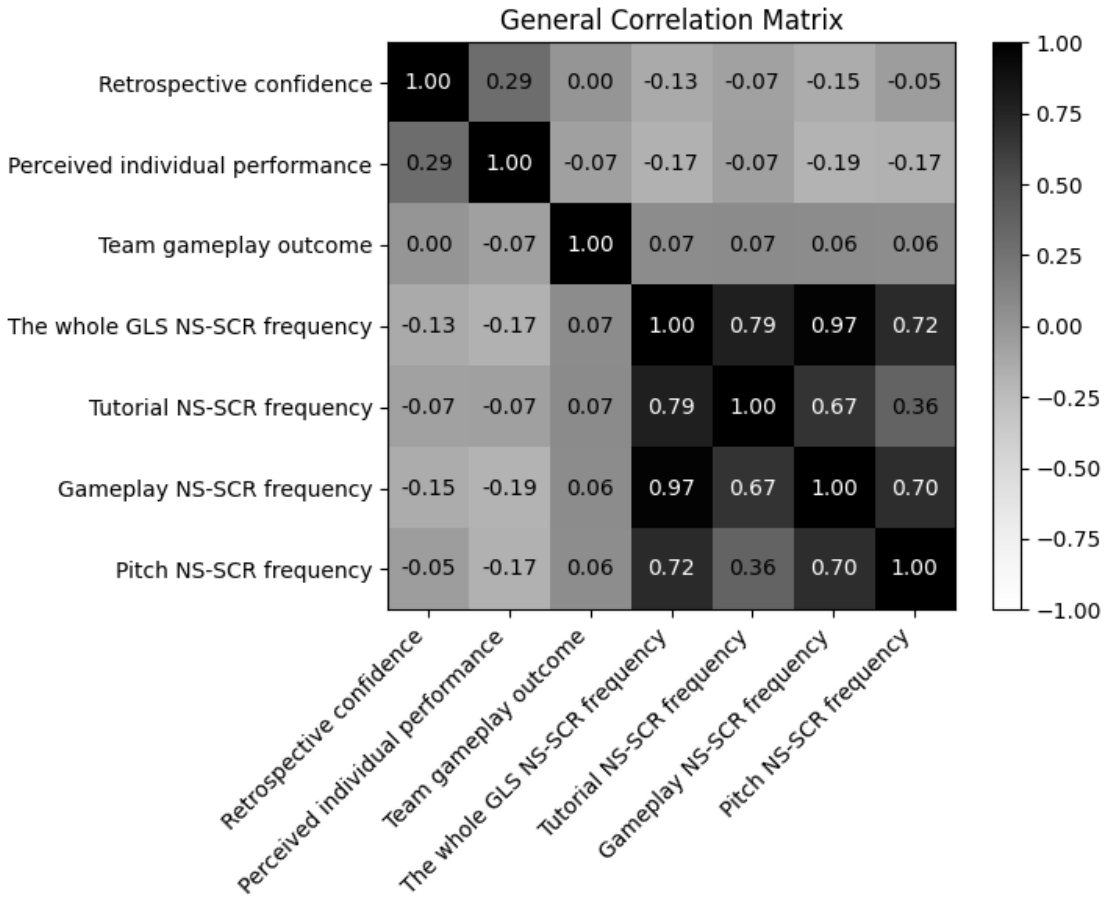
Descriptive statistics:

                          count       mean       std        min        25%  \
cf                         87.0   3.425287  1.559906   1.000000   2.000000
pp                         87.0   6.494253  1.764466   2.000000   5.000000
uncertScore                87.0  33.482759  7.603338  11.000000  30.000000
```

realGamePeaks_per_m	87.0	2.114204	2.406057	-4.070671	0.358493
realTutoPeaks_per_m	87.0	2.989865	2.548684	-3.124891	1.270447
realQuestionsPeaks_per_m	87.0	1.800861	2.490374	-4.446577	0.189259
realPitchPeaks_per_m	87.0	2.272550	3.905028	-11.221374	0.346963

	50%	75%	max
cf	3.000000	5.000000	7.000000
pp	7.000000	8.000000	10.000000
uncertScore	34.000000	37.000000	46.000000
realGamePeaks_per_m	2.079011	3.722039	8.403172
realTutoPeaks_per_m	2.378785	5.282077	8.361345
realQuestionsPeaks_per_m	1.723939	3.370871	9.234457
realPitchPeaks_per_m	2.019637	4.973874	9.502378





[68]:
```
eda_df = dat_model.copy()

eda_df[["pitch_within", "pitch_between"]] = np.nan
eda_df.loc[dat_model_pitch_role.index, ["pitch_within", "pitch_between"]] =␣
 ↪dat_model_pitch_role[["pitch_within", "pitch_between"]].to_numpy()

cols_within_between = ["cf", "pp", "uncertScore", "scr_within", "scr_between",␣
 ↪"tuto_within", "tuto_between","questions_within", "questions_between",␣
 ↪"pitch_within", "pitch_between"]
cols_within = ["cf", "pp", "uncertScore", "scr_within", "tuto_within",␣
 ↪"questions_within", "pitch_within"]
cols_between = ["cf", "pp", "uncertScore", "scr_between",  "tuto_between",␣
 ↪"questions_between", "pitch_between"]
```

Within-team Exploratory Data Analysis

[69]:
```
# Basic summary
print("Missing values:")
```

```
print(eda_df[cols_within].isna().sum(), "\n")

print("Descriptive statistics:")
display(eda_df[cols_within].describe().T)

# Distribution plots and boxplots
fig, axes = plt.subplots(2, len(cols_within), figsize=(20, 8))

for i, col in enumerate(cols_within):
    axes[0, i].hist(eda_df[col], bins=12, edgecolor="black")
    axes[0, i].set_title(f"Histogram: {col}")
    axes[0, i].set_xlabel(col)
    axes[0, i].set_ylabel("Count")

    axes[1, i].boxplot(eda_df[col], vert=True)
    axes[1, i].set_title(f"Boxplot: {col}")
    axes[1, i].set_xticks([])

plt.tight_layout()
plt.show()
```

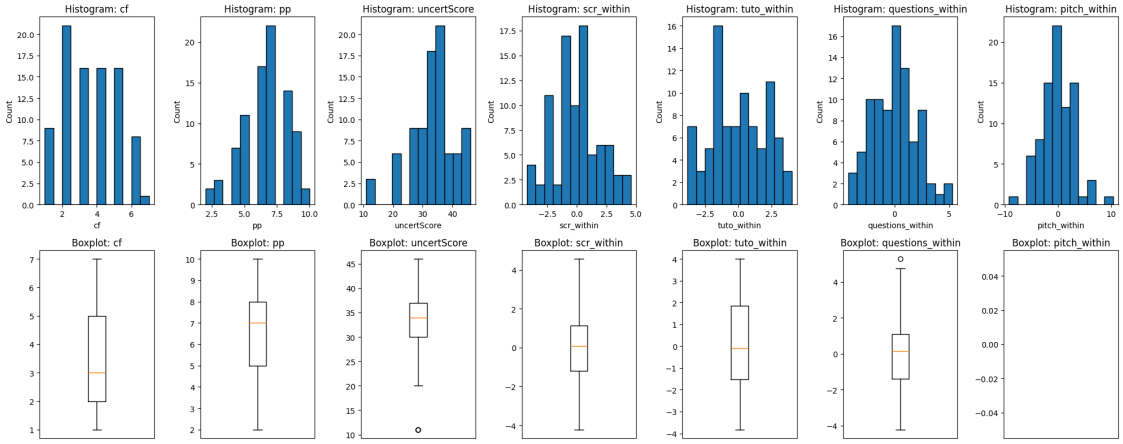
```
Missing values:
cf                  0
pp                  0
uncertScore         0
scr_within          0
tuto_within         0
questions_within    0
pitch_within        3
dtype: int64

Descriptive statistics:
```

	count	mean	std	min	25%	\
cf	87.0	3.425287e+00	1.559906	1.000000	2.000000	
pp	87.0	6.494253e+00	1.764466	2.000000	5.000000	
uncertScore	87.0	3.348276e+01	7.603338	11.000000	30.000000	
scr_within	87.0	4.083579e-17	1.952139	-4.228912	-1.186685	
tuto_within	87.0	1.020895e-17	2.049427	-3.827188	-1.513568	
questions_within	87.0	-1.786566e-17	1.992601	-4.223753	-1.400472	
pitch_within	84.0	-1.057355e-17	3.098589	-9.181356	-1.706938	

	50%	75%	max
cf	3.000000	5.000000	7.000000
pp	7.000000	8.000000	10.000000
uncertScore	34.000000	37.000000	46.000000
scr_within	0.079387	1.136715	4.572771
tuto_within	-0.096863	1.860652	4.003711

```
questions_within    0.153288    1.088501    5.281570
pitch_within       -0.176149    2.024036   10.406666
```



[70]:
```
# Correlation matrix
corr = eda_df[cols_within].corr()

fig, ax = plt.subplots(figsize=(10, 8))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)

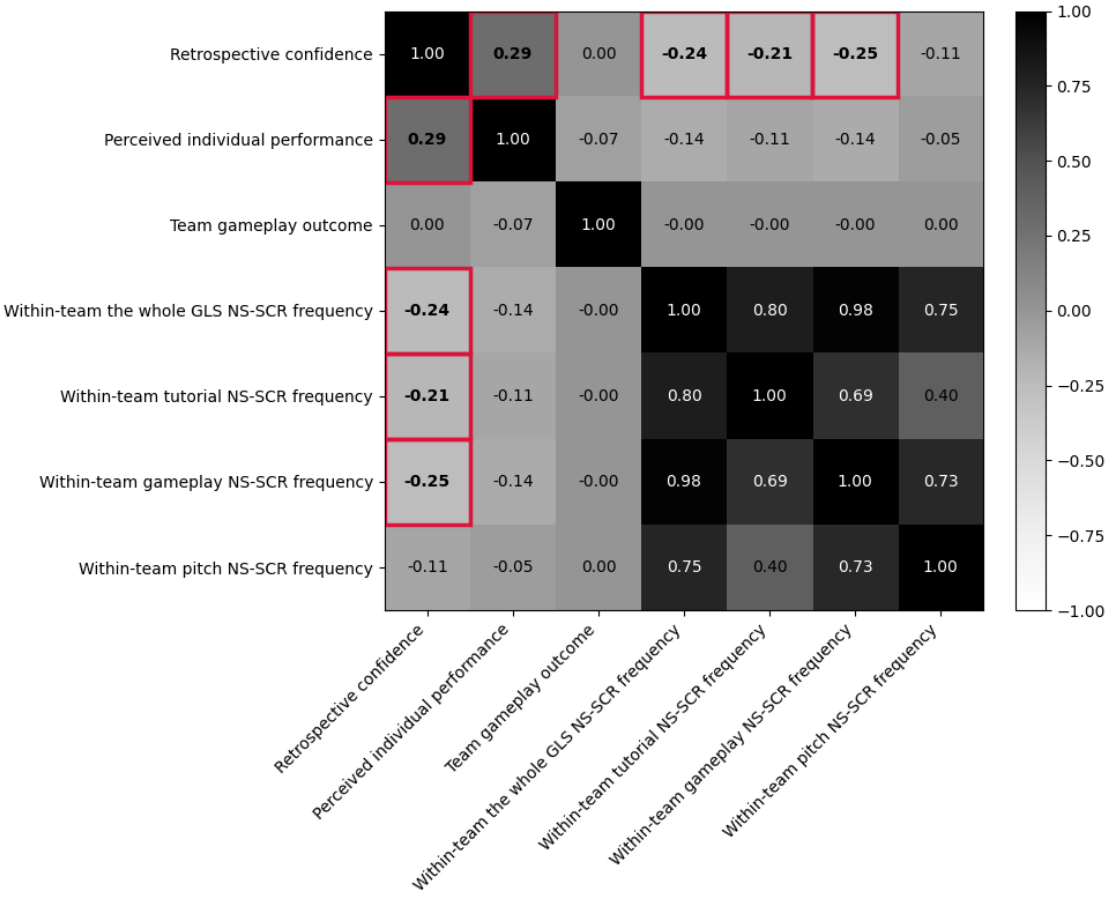
display_labels = [labels[col] for col in cols_within]

ax.set_xticks(range(len(display_labels)))
ax.set_yticks(range(len(display_labels)))
ax.set_xticklabels(display_labels, rotation=45, ha="right")
ax.set_yticklabels(display_labels)

threshold = 0.20
focus_var = "cf"

for i in range(len(cols_within)):
    for j in range(len(cols_within)):
        value = corr.iloc[i, j]
        text_color = "white" if abs(value) > 0.5 else "black"
        involves_focus = focus_var in (cols_within[i], cols_within[j])
        is_strong = abs(value) >= threshold and i != j and involves_focus
        ax.text(j, i, f"{value:.2f}", ha="center", va="center",
                color=text_color, fontweight="bold" if is_strong else "normal")
        if is_strong:
            rect = patches.Rectangle((j - 0.5, i - 0.5), 1, 1,
                                     linewidth=2.5, edgecolor="crimson",␣
 ↪facecolor="none")
            ax.add_patch(rect)
```

```
plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
plt.tight_layout()
plt.show()
```



Between-team Exploratory Data Analysis

[71]:
```
# Basic summary
print("Missing values:")
print(eda_df[cols_between].isna().sum(), "\n")

print("Descriptive statistics:")
display(eda_df[cols_between].describe().T)

# Distribution plots and boxplots
fig, axes = plt.subplots(2, len(cols_between), figsize=(20, 8))

for i, col in enumerate(cols_between):
    axes[0, i].hist(eda_df[col], bins=12, edgecolor="black")
```

```
    axes[0, i].set_title(f"Histogram: {col}")
    axes[0, i].set_xlabel(col)
    axes[0, i].set_ylabel("Count")

    axes[1, i].boxplot(eda_df[col], vert=True)
    axes[1, i].set_title(f"Boxplot: {col}")
    axes[1, i].set_xticks([])

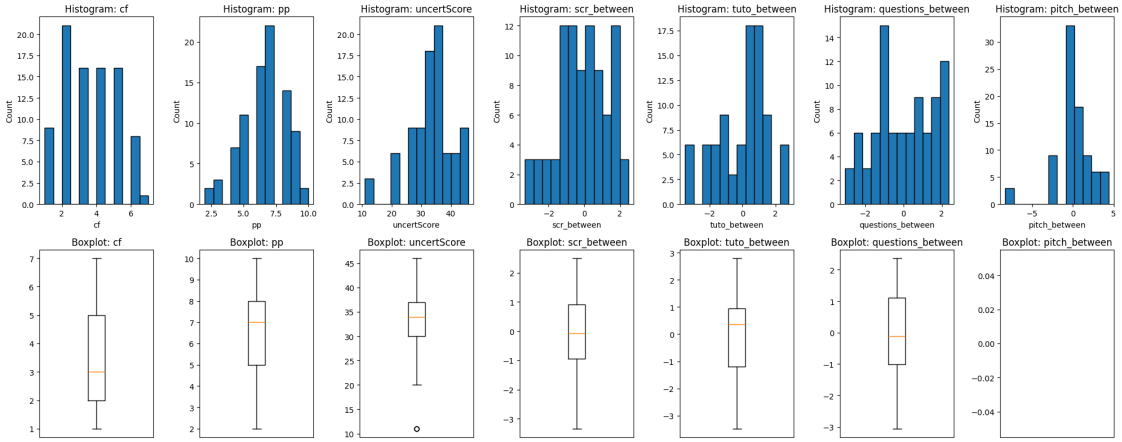
plt.tight_layout()
plt.show()
```

```
Missing values:
cf                   0
pp                   0
uncertScore          0
scr_between          0
tuto_between         0
questions_between    0
pitch_between        3
dtype: int64

Descriptive statistics:

                    count          mean       std        min        25%  \
cf                   87.0  3.425287e+00  1.559906   1.000000   2.000000
pp                   87.0  6.494253e+00  1.764466   2.000000   5.000000
uncertScore          87.0  3.348276e+01  7.603338  11.000000  30.000000
scr_between          87.0  2.756416e-16  1.406507  -3.337750  -0.941966
tuto_between         87.0 -8.473426e-16  1.515138  -3.478591  -1.187725
questions_between    87.0  8.167158e-17  1.493821  -3.056665  -1.009198
pitch_between        84.0  1.269501e-01  2.385979  -8.290591  -0.652798

                         50%        75%        max
cf                  3.000000   5.000000   7.000000
pp                  7.000000   8.000000  10.000000
uncertScore        34.000000  37.000000  46.000000
scr_between        -0.065864   0.914374   2.494181
tuto_between        0.369232   0.947484   2.792636
questions_between  -0.114830   1.106064   2.363263
pitch_between       0.139819   1.189454   4.415574
```



[72]:
```
# Correlation matrix
corr = eda_df[cols_between].corr()

fig, ax = plt.subplots(figsize=(10, 8))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)

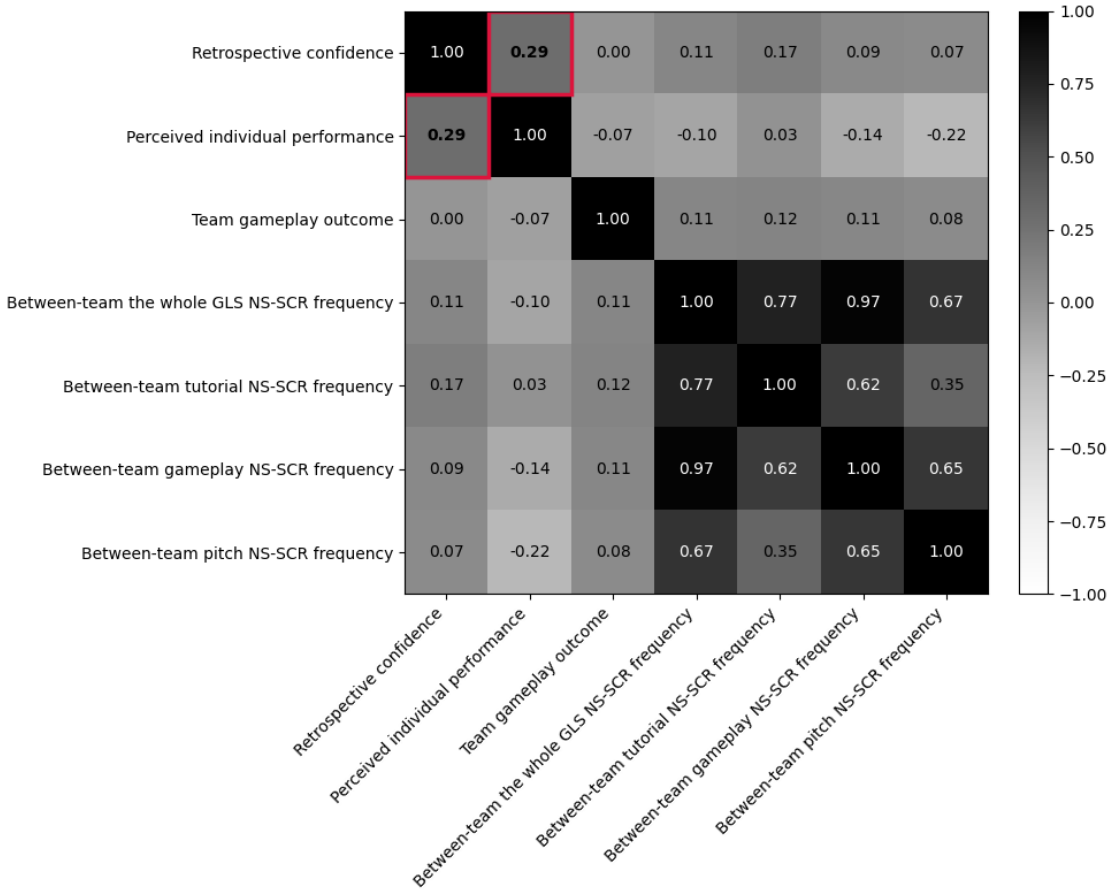
display_labels = [labels[col] for col in cols_between]

ax.set_xticks(range(len(display_labels)))
ax.set_yticks(range(len(display_labels)))
ax.set_xticklabels(display_labels, rotation=45, ha="right")
ax.set_yticklabels(display_labels)
#ax.set_title("Detailed Correlation Matrix")

threshold = 0.20
focus_var = "cf"

for i in range(len(cols_between)):
    for j in range(len(cols_between)):
        value = corr.iloc[i, j]
        text_color = "white" if abs(value) > 0.5 else "black"
        involves_focus = focus_var in (cols_between[i], cols_between[j])
        is_strong = abs(value) >= threshold and i != j and involves_focus
        ax.text(j, i, f"{value:.2f}", ha="center", va="center",
                color=text_color, fontweight="bold" if is_strong else "normal")
        if is_strong:
            rect = patches.Rectangle((j - 0.5, i - 0.5), 1, 1,
                                     linewidth=2.5, edgecolor="crimson",␣
 ↪facecolor="none")
            ax.add_patch(rect)
```

```
plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
plt.tight_layout()
plt.show()
```



[73]:
```
# Correlation matrix
corr = eda_df[cols_within_between].corr()

fig, ax = plt.subplots(figsize=(10, 8))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)

display_labels = [labels[col] for col in cols_within_between]

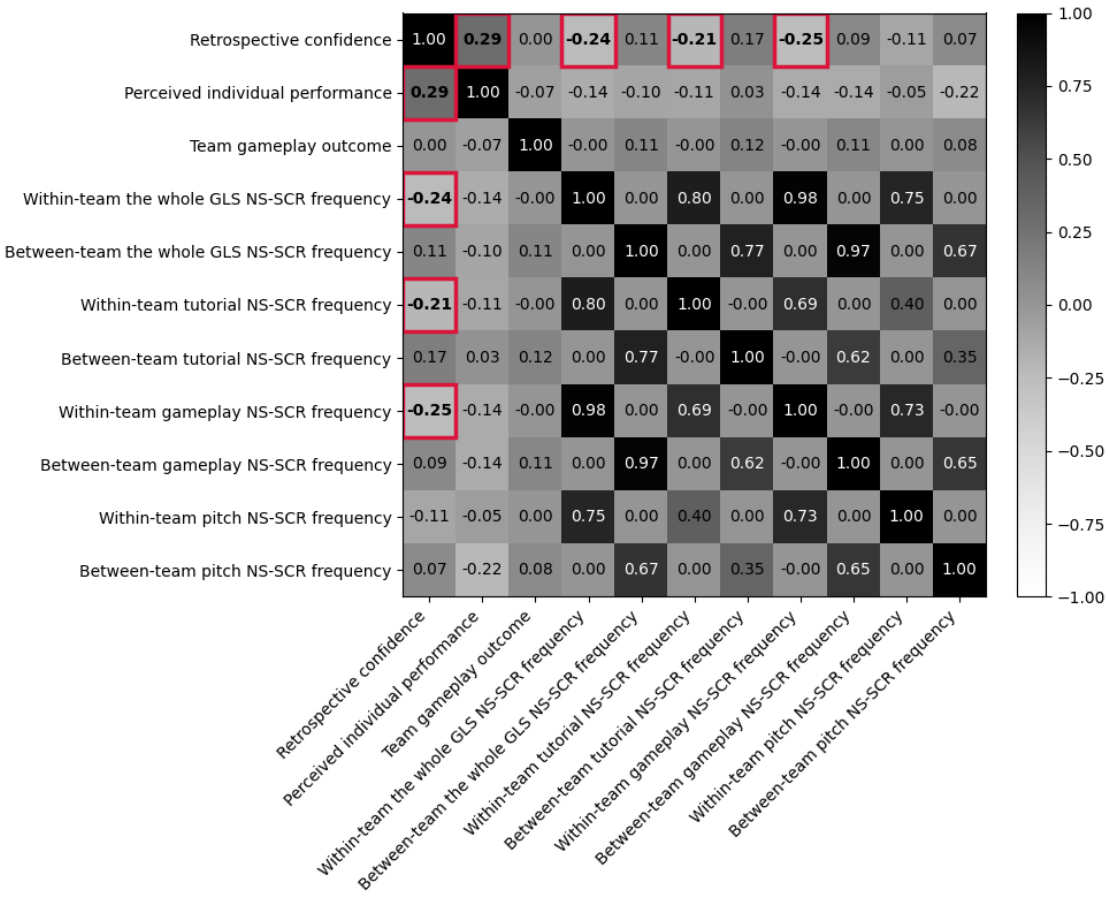
ax.set_xticks(range(len(display_labels)))
ax.set_yticks(range(len(display_labels)))
ax.set_xticklabels(display_labels, rotation=45, ha="right")
ax.set_yticklabels(display_labels)

threshold = 0.20
```

```
focus_var = "cf"

for i in range(len(cols_within_between)):
    for j in range(len(cols_within_between)):
        value = corr.iloc[i, j]
        text_color = "white" if abs(value) > 0.5 else "black"
        involves_focus = focus_var in (cols_within_between[i],␣
 ↪cols_within_between[j])
        is_strong = abs(value) >= threshold and i != j and involves_focus
        ax.text(j, i, f"{value:.2f}", ha="center", va="center",
                color=text_color, fontweight="bold" if is_strong else "normal")
        if is_strong:
            rect = patches.Rectangle((j - 0.5, i - 0.5), 1, 1,
                                     linewidth=2.5, edgecolor="crimson",␣
 ↪facecolor="none")
            ax.add_patch(rect)

plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
plt.tight_layout()
plt.show()
```



1.2.1 Model 1 - CF ~ realGamePeaks_per_m_z + (1|groupID)

1. Statsmodels (LME4)

[74]:
```
model1_sm = sm.MixedLM.from_formula(
    "cf ~ realGamePeaks_per_m_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result1_sm = model1_sm.fit(reml=True, method="lbfgs")
print("Model 1 sm: CF ~ realGamePeaks_per_m_z + (1|groupID)")
print(result1_sm.summary())
```

```
Model 1 sm: CF ~ realGamePeaks_per_m_z + (1|groupID)
             Mixed Linear Model Regression Results
================================================================
Model:               MixedLM    Dependent Variable:    cf
No. Observations:    87         Method:                REML
No. Groups:          29         Scale:                 1.8671
Min. group size:     3          Log-Likelihood:        -160.6816
Max. group size:     3          Converged:             Yes
Mean group size:     3.0
----------------------------------------------------------------
                      Coef.  Std.Err.   z    P>|z| [0.025 0.975]
----------------------------------------------------------------
Intercept              3.425    0.203 16.854 0.000  3.027  3.824
realGamePeaks_per_m_z -0.298    0.167 -1.789 0.074 -0.625  0.029
groupID Var            0.575    0.295
================================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[19]:
```
model1_bmb = bmb.Model(
    "cf ~ realGamePeaks_per_m_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model1 = model1_bmb.fit(
```

```
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 1 bmb: CF ~ realGamePeaks_per_m_z + (1|groupID)")
az.summary(idata_model1, var_names=["realGamePeaks_per_m_z", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, realGamePeaks_per_m_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 1 seconds.
There was 1 divergence after tuning. Increase `target_accept` or reparameterize.
```

```
Model 1 bmb: CF ~ realGamePeaks_per_m_z + (1|groupID)
```

[19]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	\
realGamePeaks_per_m_z	-0.293	0.174	-0.606	0.042	0.002	0.002	
1|groupID_sigma	0.727	0.277	0.149	1.217	0.007	0.004	
sigma	1.405	0.134	1.172	1.670	0.003	0.002	

	ess_bulk	ess_tail	r_hat
realGamePeaks_per_m_z	5495.0	4901.0	1.0
1|groupID_sigma	1695.0	1804.0	1.0
sigma	2684.0	3775.0	1.0

Within/Between Model 1: CF ~ scr_within_z + scr_between_z + (1|groupID)

1. Statsmodels (LME4)

[20]:
```
model1_within_sm = sm.MixedLM.from_formula(
    "cf ~ scr_within_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result1_within_sm = model1_within_sm.fit(reml=True, method="lbfgs")
print("Model 1 within sm: CF ~ scr_within_z + (1|groupID)")
print(result1_within_sm.summary())
```

```
Model 1 within sm: CF ~ scr_within_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:            MixedLM Dependent Variable: cf
No. Observations: 87      Method:             REML
No. Groups:       29      Scale:              1.8480
Min. group size:  3       Log-Likelihood:     -159.2149
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
---------------------------------------------------------
             Coef.  Std.Err.   z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept     3.425    0.194 17.632 0.000  3.045  3.806
scr_within_z -0.374    0.147 -2.548 0.011 -0.661 -0.086
groupID Var   0.478    0.263
=========================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[21]:
```
model1_between_sm = sm.MixedLM.from_formula(
    "cf ~ scr_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result1_between_sm = model1_between_sm.fit(reml=True, method="lbfgs")
print("Model 1 between sm: CF ~ scr_between_z + (1|groupID)")
print(result1_between_sm.summary())
```

```
Model 1 between sm: CF ~ scr_between_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:            MixedLM Dependent Variable: cf
No. Observations: 87      Method:             REML
No. Groups:       29      Scale:              2.0230
Min. group size:  3       Log-Likelihood:     -161.6428
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
---------------------------------------------------------
              Coef. Std.Err.   z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept     3.425    0.195 17.573 0.000  3.043  3.807
scr_between_z 0.177    0.196  0.901 0.368 -0.208  0.561
groupID Var   0.427    0.255
=========================================================
```

```
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[22]:
```
model1_within_between_sm = sm.MixedLM.from_formula(
    "cf ~ scr_within_z + scr_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result1_within_between_sm = model1_within_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 1 within-between sm: CF ~ scr_within_z + scr_between_z +␣
 ↪(1|groupID)")
print(result1_within_between_sm.summary())
```

```
Model 1 within-between sm: CF ~ scr_within_z + scr_between_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:             MixedLM Dependent Variable: cf
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              1.8480
Min. group size:   3       Log-Likelihood:     -159.5196
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
---------------------------------------------------------
              Coef.  Std.Err.   z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept      3.425    0.195 17.573 0.000  3.043  3.807
scr_within_z  -0.374    0.147 -2.548 0.011 -0.661 -0.086
scr_between_z  0.177    0.196  0.901 0.368 -0.208  0.561
groupID Var    0.486    0.268
=========================================================
```

```
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[23]:
```
model1_within_bmb  = bmb.Model(
    "cf ~ scr_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
```

```
)

idata_model1_within_bmb = model1_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 1 within bmb: CF ~ scr_within_z + (1|groupID)")
az.summary(idata_model1_within_bmb, var_names=["scr_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, scr_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 1 seconds.
There were 64 divergences after tuning. Increase `target_accept` or
reparameterize.
```

```
Model 1 within bmb: CF ~ scr_within_z + (1|groupID)
```

[23]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
scr_within_z	-0.371	0.153	-0.660	-0.085	0.002	0.002	4811.0	
1|groupID_sigma	0.644	0.272	0.059	1.098	0.007	0.004	1451.0	
sigma	1.399	0.132	1.175	1.658	0.003	0.002	2406.0	

	ess_tail	r_hat
scr_within_z	4477.0	1.0
1|groupID_sigma	1908.0	1.0
sigma	3769.0	1.0

[24]:
```
model1_between_bmb  = bmb.Model(
    "cf ~ scr_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model1_between_bmb = model1_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
```

```
    random_seed=123,
    target_accept=0.99
)

print("Model 1 between bmb: CF ~ scr_between_z + (1|groupID)")
az.summary(idata_model1_between_bmb, var_names=["scr_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, scr_between_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.
```

```
Model 1 between bmb: CF ~ scr_between_z + (1|groupID)
```

[24]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
scr_between_z	0.173	0.201	-0.207	0.545	0.003	0.002	6227.0	5718.0	1.0
1|groupID_sigma	0.611	0.279	0.012	1.041	0.007	0.003	1454.0	1929.0	1.0
sigma	1.460	0.133	1.217	1.713	0.002	0.001	3505.0	4732.0	1.0

[25]:
```
model1_within_between_bmb  = bmb.Model(
    "cf ~ scr_within_z + scr_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model1_within_between_bmb = model1_within_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 1 within-between bmb: CF ~ scr_within_z + scr_between_z +␣
 ↪(1|groupID)")
az.summary(idata_model1_within_between_bmb, var_names=["scr_within_z",␣
 ↪"scr_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, scr_within_z, scr_between_z, 1|groupID_sigma,
1|groupID_offset]

Output()


Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 4 seconds.

Model 1 within-between bmb: CF ~ scr_within_z + scr_between_z + (1|groupID)
```

[25]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
scr_within_z	-0.373	0.149	-0.647	-0.090	0.001	0.002	11983.0	
scr_between_z	0.179	0.201	-0.184	0.559	0.003	0.002	5997.0	
1|groupID_sigma	0.663	0.275	0.082	1.145	0.007	0.004	1406.0	
sigma	1.397	0.131	1.159	1.640	0.002	0.001	2930.0	

	ess_tail	r_hat
scr_within_z	5488.0	1.0
scr_between_z	5593.0	1.0
1|groupID_sigma	1239.0	1.0
sigma	5151.0	1.0

1.2.2 Model 2 - CF ~ event_phase_within/between_z + (1|groupID)

a) Tutorial

1. Statsmodels (LME4)

[26]:
```
model2_tuto_within_sm = sm.MixedLM.from_formula(
    "cf ~ tuto_within_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_tuto_within_sm = model2_tuto_within_sm.fit(reml=True, method="lbfgs")
print("Model 2 tuto within sm: CF ~ tuto_within_z + (1|groupID)")
print(result2_tuto_within_sm.summary())
```

```
Model 2 tuto within sm: CF ~ tuto_within_z + (1|groupID)
         Mixed Linear Model Regression Results
============================================================
Model:             MixedLM Dependent Variable: cf
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              1.8934
Min. group size:   3       Log-Likelihood:     -159.9074
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
```

```
------------------------------------------------------
              Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------
Intercept      3.425    0.194 17.632 0.000  3.045  3.806
tuto_within_z -0.331    0.148 -2.229 0.026 -0.622 -0.040
groupID Var    0.463    0.260
======================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[27]:
```
model2_tuto_between_sm = sm.MixedLM.from_formula(
    "cf ~ tuto_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_tuto_between_sm = model2_tuto_between_sm.fit(reml=True, method="lbfgs")
print("Model 2 tuto between sm: CF ~ tuto_between_z + (1|groupID)")
print(result2_tuto_between_sm.summary())
```

```
Model 2 tuto between sm: CF ~ tuto_between_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:             MixedLM Dependent Variable: cf
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              2.0230
Min. group size:   3       Log-Likelihood:     -161.1722
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
---------------------------------------------------------
               Coef. Std.Err.    z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept      3.425    0.192 17.882 0.000  3.050  3.801
tuto_between_z 0.258    0.193  1.341 0.180 -0.119  0.636
groupID Var    0.390    0.246
=========================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[28]:
```
model2_tuto_within_between_sm = sm.MixedLM.from_formula(
    "cf ~ tuto_within_z + tuto_between_z",
```

```
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_tuto_within_between_sm = model2_tuto_within_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 2 tuto within-between sm: CF ~ tuto_within_z + tuto_between_z +␣
 ↪(1|groupID)")
print(result2_tuto_within_between_sm.summary())
```

```
Model 2 tuto within-between sm: CF ~ tuto_within_z + tuto_between_z +
(1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: cf
No. Observations:   87      Method:             REML
No. Groups:         29      Scale:              1.8934
Min. group size:    3       Log-Likelihood:     -159.7416
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
               Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       3.425    0.192 17.882 0.000  3.050  3.801
tuto_within_z  -0.331    0.148 -2.229 0.026 -0.622 -0.040
tuto_between_z  0.258    0.193  1.341 0.180 -0.119  0.636
groupID Var     0.433    0.255
============================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[29]:
```
model2_tuto_within_bmb  = bmb.Model(
    "cf ~ tuto_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_tuto_within_bmb = model2_tuto_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
```

```
    random_seed=123,
    target_accept=0.99
)

print("Model 2 tuto within bmb: CF ~ tuto_within_z + (1|groupID)")
az.summary(idata_model2_tuto_within_bmb, var_names=["tuto_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, tuto_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.
```

```
Model 2 tuto within bmb: CF ~ tuto_within_z + (1|groupID)
```

[29]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
tuto_within_z	-0.331	0.153	-0.614	-0.038	0.001	0.002	10654.0	
1|groupID_sigma	0.631	0.277	0.049	1.095	0.008	0.004	1306.0	
sigma	1.419	0.138	1.176	1.681	0.003	0.002	2809.0	

	ess_tail	r_hat
tuto_within_z	6259.0	1.00
1|groupID_sigma	1451.0	1.01
sigma	4823.0	1.00

[30]:
```
model2_tuto_between_bmb  = bmb.Model(
    "cf ~ tuto_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_tuto_between_bmb = model2_tuto_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 tuto between bmb: CF ~ tuto_between_z + (1|groupID)")
az.summary(idata_model2_tuto_between_bmb, var_names=["tuto_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, tuto_between_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 4 seconds.
```

```
Model 2 tuto between bmb: CF ~ tuto_between_z + (1|groupID)
```

[30]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
tuto_between_z	0.257	0.196	-0.102	0.642	0.003	0.003	5969.0	5659.0	1.0
1|groupID_sigma	0.582	0.283	0.002	1.017	0.007	0.003	1406.0	1613.0	1.0
sigma	1.458	0.134	1.213	1.710	0.002	0.001	3579.0	5319.0	1.0

[31]:
```
model2_tuto_within_between_bmb  = bmb.Model(
    "cf ~ tuto_within_z + tuto_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_tuto_within_between_bmb = model2_tuto_within_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 tuto within-between bmb: CF ~ tuto_within_z + tuto_between_z +␣
 ↪(1|groupID)")
az.summary(idata_model2_tuto_within_between_bmb, var_names=["tuto_within_z",␣
 ↪"tuto_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, tuto_within_z, tuto_between_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 4 seconds.

Model 2 tuto within-between bmb: CF ~ tuto_within_z + tuto_between_z +
(1|groupID)
```

[31]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
tuto_within_z	-0.331	0.154	-0.618	-0.042	0.001	0.002	11871.0	
tuto_between_z	0.259	0.197	-0.116	0.630	0.003	0.002	6119.0	
1|groupID_sigma	0.615	0.276	0.018	1.046	0.008	0.004	1324.0	
sigma	1.414	0.133	1.170	1.658	0.002	0.001	3106.0	

	ess_tail	r_hat
tuto_within_z	5929.0	1.0
tuto_between_z	5801.0	1.0
1|groupID_sigma	1400.0	1.0
sigma	5239.0	1.0

b) Questions

1. Statsmodels (LME4)

[32]:
```
model2_questions_within_sm = sm.MixedLM.from_formula(
    "cf ~ questions_within_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_questions_within_sm = model2_questions_within_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 2 questions within sm: CF ~ questions_within_z + (1|groupID)")
print(result2_questions_within_sm.summary())
```

```
Model 2 questions within sm: CF ~ questions_within_z + (1|groupID)
            Mixed Linear Model Regression Results
============================================================
Model:              MixedLM   Dependent Variable:   cf
No. Observations:   87        Method:               REML
No. Groups:         29        Scale:                1.8232
Min. group size:    3         Log-Likelihood:       -158.8308
Max. group size:    3         Converged:            Yes
Mean group size:    3.0
------------------------------------------------------------
                   Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept           3.425    0.194 17.632 0.000  3.045  3.806
questions_within_z -0.395    0.146 -2.712 0.007 -0.680 -0.109
groupID Var         0.487    0.265
============================================================
```

```
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[33]:
```
model2_questions_between_sm = sm.MixedLM.from_formula(
    "cf ~ questions_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_questions_between_sm = model2_questions_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 2 questions between sm: CF ~ questions_between_z + (1|groupID)")
print(result2_questions_between_sm.summary())
```

```
Model 2 questions between sm: CF ~ questions_between_z + (1|groupID)
            Mixed Linear Model Regression Results
============================================================
Model:              MixedLM   Dependent Variable:   cf
No. Observations:   87        Method:               REML
No. Groups:         29        Scale:                2.0230
Min. group size:    3         Log-Likelihood:       -161.7841
Max. group size:    3         Converged:            Yes
Mean group size:    3.0
------------------------------------------------------------
                    Coef. Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept           3.425    0.196 17.481 0.000  3.041  3.809
questions_between_z 0.142    0.197  0.723 0.470 -0.244  0.529
groupID Var         0.439    0.258
============================================================
```

```
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[34]:
```
model2_questions_within_between_sm = sm.MixedLM.from_formula(
    "cf ~ questions_within_z + questions_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_questions_within_between_sm = model2_questions_within_between_sm.
 ↪fit(reml=True, method="lbfgs")
```

```
print("Model 2 questions within-between sm: CF ~ questions_within_z +␣
 ↪questions_between_z + (1|groupID)")
print(result2_questions_within_between_sm.summary())
```

```
Model 2 questions within-between sm: CF ~ questions_within_z +
questions_between_z + (1|groupID)
            Mixed Linear Model Regression Results
============================================================
Model:               MixedLM   Dependent Variable:   cf
No. Observations:    87        Method:               REML
No. Groups:          29        Scale:                1.8232
Min. group size:     3         Log-Likelihood:       -159.2768
Max. group size:     3         Converged:            Yes
Mean group size:     3.0
------------------------------------------------------------
                    Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept            3.425    0.196 17.481 0.000  3.041  3.809
questions_within_z  -0.395    0.146 -2.712 0.007 -0.680 -0.110
questions_between_z  0.142    0.197  0.723 0.470 -0.244  0.529
groupID Var          0.506    0.272
============================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[35]:
```
model2_questions_within_bmb  = bmb.Model(
    "cf ~ questions_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_questions_within_bmb = model2_questions_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 2 questions within bmb: CF ~ questions_within_z + (1|groupID)")
az.summary(idata_model2_questions_within_bmb, var_names=["questions_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, questions_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There were 52 divergences after tuning. Increase `target_accept` or
reparameterize.
```

```
Model 2 questions within bmb: CF ~ questions_within_z + (1|groupID)
```

[35]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	\
questions_within_z	-0.396	0.152	-0.697	-0.123	0.002	0.003	
1|groupID_sigma	0.659	0.267	0.119	1.144	0.008	0.004	
sigma	1.389	0.131	1.147	1.625	0.003	0.002	

	ess_bulk	ess_tail	r_hat
questions_within_z	4089.0	2600.0	1.0
1|groupID_sigma	1099.0	2147.0	1.0
sigma	1587.0	1906.0	1.0

[36]:
```
model2_questions_between_bmb  = bmb.Model(
    "cf ~ questions_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_questions_between_bmb = model2_questions_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 questions between bmb: CF ~ questions_between_z + (1|groupID)")
az.summary(idata_model2_questions_between_bmb,␣
 ↪var_names=["questions_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, questions_between_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.

Model 2 questions between bmb: CF ~ questions_between_z + (1|groupID)
```

[36]:
```
                       mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  \
questions_between_z   0.141  0.200  -0.237    0.513      0.003    0.002
1|groupID_sigma       0.619  0.283   0.000    1.044      0.007    0.004
sigma                 1.458  0.134   1.213    1.709      0.002    0.002

                      ess_bulk  ess_tail  r_hat
questions_between_z     5297.0    5488.0    1.0
1|groupID_sigma         1511.0    1691.0    1.0
sigma                   3060.0    4972.0    1.0
```

[37]:
```
model2_questions_within_between_bmb  = bmb.Model(
    "cf ~ questions_within_z + questions_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_questions_within_between_bmb = model2_questions_within_between_bmb.
 ↪fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 questions within-between bmb: CF ~ questions_within_z +␣
 ↪questions_between_z + (1|groupID)")
az.summary(idata_model2_questions_within_between_bmb,␣
 ↪var_names=["questions_within_z", "questions_between_z", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, questions_within_z, questions_between_z,
1|groupID_sigma, 1|groupID_offset]

Output()


Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 4 seconds.

Model 2 questions within-between bmb: CF ~ questions_within_z +
questions_between_z + (1|groupID)
```

[37]:
```
                       mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  \
questions_within_z   -0.395  0.150  -0.677   -0.105      0.001    0.002
questions_between_z   0.141  0.203  -0.264    0.501      0.003    0.002
1|groupID_sigma       0.675  0.272   0.079    1.135      0.007    0.004
sigma                 1.387  0.131   1.142    1.627      0.002    0.001

                     ess_bulk  ess_tail  r_hat
questions_within_z    12271.0    6170.0    1.0
questions_between_z    4990.0    4878.0    1.0
1|groupID_sigma        1467.0    1504.0    1.0
sigma                  3354.0    4823.0    1.0
```

c) Pitch

1. Statsmodels (LME4)

[38]:
```
model2_pitch_within_sm = sm.MixedLM.from_formula(
    "cf ~ pitch_within_z",
    groups="groupID",
    re_formula="1",
    data=dat_model_pitch_role
)
result2_pitch_within_sm = model2_pitch_within_sm.fit(reml=True, method="lbfgs")
print("Model 2 pitch within sm: CF ~ pitch_within_z + (1|groupID)")
print(result2_pitch_within_sm.summary())
```

```
Model 2 pitch within sm: CF ~ pitch_within_z + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: cf
No. Observations:   84      Method:             REML
No. Groups:         28      Scale:              1.9327
Min. group size:    3       Log-Likelihood:     -155.4275
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
               Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       3.417    0.201 16.987 0.000  3.022  3.811
pitch_within_z -0.166    0.150 -1.107 0.268 -0.461  0.128
groupID Var     0.488    0.271
============================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[39]:
```
model2_pitch_between_sm = sm.MixedLM.from_formula(
    "cf ~ pitch_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model_pitch_role
)
result2_pitch_between_sm = model2_pitch_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 2 pitch between sm: CF ~ pitch_between_z + (1|groupID)")
print(result2_pitch_between_sm.summary())
```

```
Model 2 pitch between sm: CF ~ pitch_between_z + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: cf
No. Observations:   84      Method:             REML
No. Groups:         28      Scale:              1.9405
Min. group size:    3       Log-Likelihood:     -155.5600
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
                Coef. Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       3.411    0.204 16.713 0.000  3.011  3.811
pitch_between_z 0.115    0.210  0.550 0.582 -0.296  0.526
groupID Var     0.516    0.280
============================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[40]:
```
model2_pitch_within_between_sm = sm.MixedLM.from_formula(
    "cf ~ pitch_within_z + pitch_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model_pitch_role
)
result2_pitch_within_between_sm = model2_pitch_within_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 2 pitch within-between sm: CF ~ pitch_within_z + pitch_between_z +␣
 ↪(1|groupID)")
print(result2_pitch_within_between_sm.summary())
```

```
Model 2 pitch within-between sm: CF ~ pitch_within_z + pitch_between_z +
(1|groupID)
          Mixed Linear Model Regression Results
```

```
============================================================
Model:              MixedLM  Dependent Variable:  cf
No. Observations:   84       Method:              REML
No. Groups:         28       Scale:               1.9327
Min. group size:    3        Log-Likelihood:      -155.9245
Max. group size:    3        Converged:           Yes
Mean group size:    3.0
------------------------------------------------------------
                Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept        3.411    0.204 16.713 0.000  3.011  3.811
pitch_within_z  -0.166    0.150 -1.107 0.268 -0.461  0.128
pitch_between_z  0.115    0.210  0.550 0.582 -0.296  0.526
groupID Var      0.518    0.281
============================================================
```

```
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[41]:
```
model2_pitch_within_bmb  = bmb.Model(
    "cf ~ pitch_within_z + (1|groupID)",
    data=dat_model_pitch_role,
    family="gaussian"
)

idata_model2_pitch_within_bmb = model2_pitch_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 2 pitch within bmb: CF ~ pitch_within_z + (1|groupID)")
az.summary(idata_model2_pitch_within_bmb, var_names=["pitch_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pitch_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
```

```
total) took 2 seconds.
There were 187 divergences after tuning. Increase `target_accept` or
reparameterize.

Model 2 pitch within bmb: CF ~ pitch_within_z + (1|groupID)
```

[41]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
pitch_within_z	-0.166	0.156	-0.457	0.137	0.002	0.002	5687.0	4425.0	1.0
1|groupID_sigma	0.655	0.279	0.071	1.128	0.008	0.003	1315.0	2591.0	1.0
sigma	1.425	0.144	1.169	1.700	0.004	0.003	1159.0	576.0	1.0

[42]:
```
model2_pitch_between_bmb  = bmb.Model(
    "cf ~ pitch_between_z + (1|groupID)",
    data=dat_model_pitch_role,
    family="gaussian"
)

idata_model2_pitch_between_bmb = model2_pitch_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 pitch between bmb: CF ~ pitch_between_z + (1|groupID)")
az.summary(idata_model2_pitch_between_bmb, var_names=["pitch_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pitch_between_z, 1|groupID_sigma, 1|groupID_offset]

Output()


Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.

Model 2 pitch between bmb: CF ~ pitch_between_z + (1|groupID)
```

[42]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk
pitch_between_z	0.112	0.219	-0.290	0.529	0.003	0.002	5771.0
1|groupID_sigma	0.675	0.291	0.074	1.182	0.008	0.004	1434.0

```
sigma            1.435  0.138   1.184    1.690      0.003    0.002     2902.0

                 ess_tail  r_hat
pitch_between_z    6026.0    1.0
1|groupID_sigma    1321.0    1.0
sigma              4797.0    1.0
```

[43]:
```
model2_pitch_within_between_bmb  = bmb.Model(
    "cf ~ pitch_within_z + pitch_between_z + (1|groupID)",
    data=dat_model_pitch_role,
    family="gaussian"
)

idata_model2_pitch_within_between_bmb = model2_pitch_within_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 2 pitch within-between bmb: CF ~ pitch_within_z + pitch_between_z␣
 ↪+ (1|groupID)")
az.summary(idata_model2_pitch_within_between_bmb, var_names=["pitch_within_z",␣
 ↪"pitch_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pitch_within_z, pitch_between_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There was 1 divergence after tuning. Increase `target_accept` or reparameterize.
```

```
Model 2 pitch within-between bmb: CF ~ pitch_within_z + pitch_between_z +
(1|groupID)
```

[43]:
```
                  mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  \
pitch_within_z  -0.165  0.155  -0.462    0.118      0.002    0.002    8711.0
pitch_between_z  0.116  0.220  -0.322    0.506      0.003    0.003    4343.0
1|groupID_sigma  0.676  0.289   0.000    1.108      0.008    0.004    1337.0
sigma            1.433  0.139   1.184    1.693      0.003    0.001    2777.0

                 ess_tail  r_hat
pitch_within_z     5469.0    1.0
```

```
pitch_between_z     3655.0    1.0
1|groupID_sigma     1748.0    1.0
sigma               4716.0    1.0
```

1.2.3 Model 3 - PP ~ cf_z + (1|groupID)

1. Statsmodels (LME4)

[75]:
```
model3_0_sm = sm.MixedLM.from_formula(
    "pp ~ cf_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result3_0_sm = model3_0_sm.fit(reml=True, method="lbfgs")
print("Model 3_0 sm: PP ~ CF_z + (1|groupID)")
print(result3_0_sm.summary())
```

```
Model 3_0 sm: PP ~ CF_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:             MixedLM Dependent Variable: pp
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              2.5409
Min. group size:   3       Log-Likelihood:     -169.5847
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
---------------------------------------------------------
              Coef. Std.Err.   z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept     6.494    0.204 31.839 0.000  6.094  6.894
cf_z          0.428    0.201  2.127 0.033  0.034  0.823
groupID Var   0.360    0.269
=========================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[51]:
```
model3_0_bmb  = bmb.Model(
    "pp ~ cf_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)
```

```
idata_model3_0_bmb = model3_0_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 3_0 bmb: PP ~ cf_z + (1|groupID)")
az.summary(idata_model3_0_bmb, var_names=["cf_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, cf_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 4 seconds.
```

```
Model 3_0 bmb: PP ~ cf_z + (1|groupID)
```

[51]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
cf_z	0.427	0.198	0.065	0.809	0.003	0.002	5547.0	
1|groupID_sigma	0.574	0.308	0.001	1.057	0.008	0.003	1467.0	
sigma	1.621	0.146	1.341	1.885	0.003	0.002	3153.0	

	ess_tail	r_hat
cf_z	5851.0	1.0
1|groupID_sigma	2437.0	1.0
sigma	5076.0	1.0

1.2.4 Model 4 - uncertScore ~ cf_z + (1|groupID)

1. Statsmodels (LME4)

[89]:
```
# Model 4: team-level OLS (no groups, no random effect)
result14_ols = smf.ols("uncertScore ~ cf_between_z", data=team_model).fit()
print(result14_ols.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:            uncertScore   R-squared:                       0.000
Model:                            OLS   Adj. R-squared:                 -0.037
Method:                 Least Squares   F-statistic:                0.0005817
Date:                Thu, 01 Oct 2026   Prob (F-statistic):              0.981
Time:                        12:12:30   Log-Likelihood:                -99.810
No. Observations:                  29   AIC:                             203.6
```

```
Df Residuals:                       27   BIC:                             206.4
Df Model:                            1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      33.4828      1.455     23.015      0.000      30.498      36.468
cf_between_z    0.0353      1.463      0.024      0.981      -2.967       3.038
==============================================================================
Omnibus:                        5.979   Durbin-Watson:                   1.552
Prob(Omnibus):                  0.050   Jarque-Bera (JB):                4.186
Skew:                          -0.755   Prob(JB):                        0.123
Kurtosis:                       4.087   Cond. No.                         1.01
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
```

2. Bambi (Robustness testing)

[88]:
```
model4_bmb  = bmb.Model(
    "uncertScore ~ cf_between_z",
    data=team_model,
    family="gaussian"
)

idata_model4_bmb = model4_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 4 bmb: UncertScore ~ cf_between_z + (1|groupID)")
az.summary(idata_model4_bmb, var_names=["cf_between_z", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, cf_between_z]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 1 seconds.
```

```
Model 4 bmb: UncertScore ~ cf_between_z + (1|groupID)
```

[88]:
```
                mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  \
cf_between_z   0.055  1.529  -3.120    2.691      0.020    0.020    5622.0
sigma          8.008  1.124   6.102   10.157      0.018    0.017    4149.0

               ess_tail  r_hat
cf_between_z     4405.0    1.0
sigma            4308.0    1.0
```

1.2.5 Follow-up models

Model Follow-up 1 CF ~ questionPath + (1|groupID)

1. Statsmodels (LME4)

[59]:
```
modelFU1_sm = sm.MixedLM.from_formula(
    "cf ~ questionPath",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
resultFU1_sm = modelFU1_sm.fit(reml=True, method="lbfgs")
print("Model FU1 sm: CF ~ questionPath + (1|groupID)")
print(resultFU1_sm.summary())
```

```
Model FU1 sm: CF ~ questionPath + (1|groupID)
           Mixed Linear Model Regression Results
============================================================
Model:               MixedLM  Dependent Variable:  cf
No. Observations:    87       Method:              REML
No. Groups:          29       Scale:               2.0230
Min. group size:     3        Log-Likelihood:      -161.1184
Max. group size:     3        Converged:           Yes
Mean group size:     3.0
------------------------------------------------------------
                  Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept          3.500    0.237 14.779 0.000  3.036  3.964
questionPath[T.1] -0.241    0.425 -0.566 0.571 -1.074  0.592
groupID Var        0.447    0.260
============================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[60]:
```
modelFU1_bmb  = bmb.Model(
    "cf ~ questionPath + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_modelFU1_bmb = modelFU1_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model FU1 bmb: CF ~ questionPath + (1|groupID)")
az.summary(idata_modelFU1_bmb, var_names=["questionPath", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, questionPath, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
```

```
Model FU1 bmb: CF ~ questionPath + (1|groupID)
```

[60]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
questionPath[1]	-0.25	0.439	-0.95	0.45	6070	5460	1.00	
1|groupID_sigma	0.63	0.29	0.13	1.1	1094	967	1.00	
sigma	1.458	0.134	1.3	1.7	3191	4209	1.00	

	mcse_mean	mcse_sd
questionPath[1]	0.0056	0.0041
1|groupID_sigma	0.0086	0.0058
sigma	0.0024	0.0017

Model Follow-up 2 CF ~ pitchRole + (1|groupID)

1. Statsmodels (LME4)

[61]:
```
modelFU2_sm = sm.MixedLM.from_formula(
    "cf ~ pitchRole",
    groups="groupID",
    re_formula="1",
```

```
    data=dat_model_pitch_role
)
resultFU2_sm = modelFU2_sm.fit(reml=True, method="lbfgs")
print("Model FU2 sm: CF ~ pitchRole + (1|groupID)")
print(resultFU2_sm.summary())
```

```
Model FU2 sm: CF ~ pitchRole + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: cf
No. Observations:   84      Method:             REML
No. Groups:         28      Scale:              1.8873
Min. group size:    3       Log-Likelihood:     -153.6994
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
               Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       3.735    0.268 13.937 0.000  3.210  4.260
pitchRole[T.1] -0.581    0.328 -1.774 0.076 -1.224  0.061
groupID Var     0.480    0.268
============================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[62]:
```
modelFU2_bmb  = bmb.Model(
    "cf ~ pitchRole + (1|groupID)",
    data=dat_model_pitch_role,
    family="gaussian"
)

idata_modelFU2_bmb = modelFU2_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model FU2 bmb: CF ~ pitchRole + (1|groupID)")
az.summary(idata_modelFU2_bmb, var_names=["pitchRole", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pitchRole, 1|groupID_sigma, 1|groupID_offset]

Output()


Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 1 seconds.

Model FU2 bmb: CF ~ pitchRole + (1|groupID)
```

[62]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
pitchRole[1]	-0.589	0.338	-1.1	-0.053	7342	5596	1.00	
1|groupID_sigma	0.66	0.28	0.17	1.1	1506	1845	1.00	
sigma	1.408	0.137	1.2	1.6	2715	3478	1.00	

	mcse_mean	mcse_sd
pitchRole[1]	0.0039	0.0028
1|groupID_sigma	0.0072	0.005
sigma	0.0026	0.002